

HYPERACTIVITY OF ANTERIOR PITUITARY

1. GIGANTISM

Gigantism is the disorder characterized by excess growth of body before epiphysial fusion.

Cause :—

1. **Gigantism is due to hypersecretion of GH** in childhood or pre-adult life.
2. **Hypersecretion of GH is because of tumor** of Anterior pituitary [Acidophil cells].

Signs and Symptoms :—

1. **General overgrowth of body:**
 - $\hookrightarrow$ Huge stature with height more than 7 or 8 feet.
2. **Limbs disproportionately long**
3. **Giants are hyperglycemic:**
 - $\downarrow$ *develop*
 - Glycosuria & pituitary diabetes.
4. **Constant headache**
5. **Visual disturbances:**
 - $\hookrightarrow$ Leads to **bitemporal hemianopia**.

2. ACROMEGALY

- **Disorder characterised by:** Enlargement, thickening, and broadening of bones, particularly in extremities of body.

CAUSES

- **Due to hypersecretion of GH in adults** after fusion of epiphysis with shaft of bone.
- **Hypersecretion of GH is becoz of tumor** of acidophil cells in the anterior pituitary.

SIGNS AND SYMPTOMS

1. **Gorilla Face** —→ Face with rough features:
 - i. Protrusion of supraorbital ridges
 - ii. Broadening of nose
 - iii. Thickening of lips
 - iv. Thickening and wrinkles formation on forehead
 - v. Prognathism (*protrusion of lower jaw*)
2. **Bulldog Scalp** —→ Thickening of scalp, scalp thrown into folds / wrinkles.
3. **Bitemporal Hemianopia** —→ Visual disturbance
4. **Headache**
5. **Kyphosis** (*Extreme curvature of upper back – Thoracic spine*)
6. **Overgrowth of body hair.**
7. **Enlargement of hands and feet.**
8. **Enlargement of visceral organs** such as lungs, thymus, heart, liver and spleen.
9. **Hyperactivity of thyroid, parathyroid and adrenal glands.**
10. **Hyperglycemia and glycosuria**, resulting in **diabetes mellitus**.
11. **Hypertension.**

3. ACROMEGALIC GIGANTISM

- **A rare disorder** with symptoms of both **gigantism** and **acromegaly**.
- **Hypersecretion of GH in children**, before fusion of epiphysis with shaft of bones:
 - ↓ *causes*
 - **Gigantism**
- **Hypersecretion of GH continued** even after fusion of epiphysis:
 - ↓
 - **Symptoms of acromegaly also appear.**

HYPOACTIVITY OF ANTERIOR PITUITARY

DWARFISM

- **Pituitary disorder in children**, characterised by **stunted growth**.

CAUSES

- **Reduction in GH secretion** in infancy / early childhood causes dwarfism.
 - i. **Tumor of chromophobes** (*non-functioning tumor, compresses & destroys the normal cells secreting GH*)
 - ii. **Deficiency of GH-releasing hormone (GHRH)** secreted by hypothalamus.
 - iii. **Deficiency of Somatomedin C**
 - iv. **Atrophy or degeneration of acidophilic cells** in anterior pituitary.
 - v. **Panhypopituitarism** → Reduction in secretion of all hormones of anterior pituitary gland.

SIGNS AND SYMPTOMS

1. **Stunted skeletal growth** (*Height is only about 3 feet*)
2. **Head becomes slightly larger** in relation to body.
3. **Normal mental activity** with **no** mental retardation.
4. **Reproduction functions are normal**.
5. **During panhypopituitarism**, dwarfs do not attain puberty due to deficiency of gonadotropic hormones.

OTHER TYPES OF DWARFISM

1. LARON DWARFISM

- **Genetic disorder**
- Also called **GH-insensitivity**

- ↓
- Due to presence of abnormal GH secretagogue receptor in liver (GHS)
- ↓
- Because of mutations of genes for receptors.

2. PSYCHOGENIC DWARFISM

- Child exposed to **extreme emotional deprivation or stress**
- **Short stature** (*deficiency of GH*)
- Also called **psychosocial dwarfism / stress dwarfism**.

3. DWARFISM IN DYSTROPHIA ADIPOSEGENITALIS

- Also called **Fröhlich syndrome**
 - ↓
 - Pituitary disorder

4. DWARFISM IN PANHYPOPITUITARISM

- **Pituitary disorder** due to reduction in secretion of **all anterior pituitary hormones**.
- **Dwarfs do not attain puberty**.